

Undergraduate Computer Science student at Binus University specializing in Intelligent Systems, with a strong mathematical foundation and a passion for lower-level mechanics, algorithmic design, and autonomous decision-making systems. Practical experience spanning quantitative trading systems, low-level OS development, machine learning vision pipelines, and full-stack software development. Seeking opportunities to apply expertise in AI, graphics, robotics (UAVs), and quantitative finance.

Technical Skills

- 🔗 **Languages:** C++, C#, Python, Java, x86 Assembly, Lean 4 (Theorem Proving)
- 🔗 **AI & Machine Learning:** PyTorch, Pandas, NumPy, OpenCV, MediaPipe
- 🔗 **Game Development & Computer Graphics:** Unity (C#), HLSL/GLSL Shader Fundamentals, Graphics Programming
- 🔗 **Systems & Tools:** Linux/Unix Environments, Git/GitHub, Low-Level Kernel Programming
- 🔗 **Domain Focus:** Intelligent Systems, Quantitative Finance & Trading Algorithms, UAV Robotics, Game Development

Projects

OpenTrade — Personal Quantitative Trading System

Developed an autonomous financial decision-making agent leveraging Intelligent Systems principles.:

- 🔗 Designed and built an automated trading environment capable of processing real-time and historical financial datasets to evaluate market conditions.
- 🔗 Implemented decision-making algorithms to dynamically determine **Buy**, **Hold**, or **Sell** signals without human intervention.
- 🔗 Integrated numerical processing libraries (**Pandas**, **NumPy**, **PyTorch**) to optimize model inference and historical backtesting execution.

MakerOS — Custom Low-Level Operating System

An experimental OS built to explore low-level hardware abstraction and game execution environments.

- 🔗 Programmed foundational kernel and system modules using **Assembly (x86)** and **C**.
- 🔗 Implemented memory management and basic I/O interfaces from scratch to understand hardware-level execution.
- 🔗 Designed the OS framework with the intent to run a custom low-level game engine and lightweight graphics pipeline.

An interactive computer vision application for real-time air-writing and Korean character recognition.

- ▣ Implemented a hand-tracking pipeline using **MediaPipe** to capture user gestures via real-time camera input on a digital canvas interface.
- ▣ Trained and integrated a Machine Learning model using **PyTorch** and **OpenCV** to classify hand-drawn Korean (Hangul) characters accurately.
- ▣ Optimized frame-by-frame image preprocessing for smooth gesture tracking and low-latency predictions.

Personal Finance Management System — Front-End Developer

A smart personal tracking application designed to help users manage budget allocations and spending patterns.

- ▣ Designed and developed an intuitive user interface (UI) to display spending analytics and financial summaries.
- ▣ Built automated tracking routines to organize user transaction data and encourage disciplined spending behavior.

Education

2021–2024

Diponegoro Highschool

Blitar, Indonesia

2024–2028

Binus University

Malang, Indonesia

School Of Computer Science

